

Merton Jump-Diffusion:
Pricing and Harvesting the Equity Jump Risk Premium
Stochastic Processes — Final Group Project

Group 3

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Two empirical failures of B&S:

- **Fat tails / leptokurtosis:** returns have heavier tails than the normal distribution
- **Volatility smile:** implied vols are not flat across strikes — deep OTM options are systematically underpriced

Root cause: stock prices **jump** discontinuously in response to news, earnings surprises, and macro shocks.

Literature Map

Model	Key contribution
Merton (1976)	First jump-diffusion
Kou (2002)	Asymmetric jumps
Carr & Wu (2003)	Empirical test
Eraker et al. (2003)	Volatility jumps
Hawkes (1971)	Self-exciting intensity
Agram et al. (2024)	Deep-learning hedge

Merton (1976): The First Jump-Diffusion Model

Price dynamics

$$\frac{dS_t}{S_t} = (\alpha - \lambda k) dt + \sigma dZ_t + dq_t$$

- dZ_t : Brownian diffusion; dq_t : Poisson jump, intensity λ
- $k = \mathbb{E}(Y - 1)$: mean jump size

Option pricing formula

$$F = \sum_{n=0}^{\infty} \frac{e^{-\lambda\tau} (\lambda\tau)^n}{n!} W\left(S, \tau; E, \sigma^2 + \frac{n\delta^2}{\tau}, r\right)$$

Poisson-weighted average of B&S prices.

Key insight

Jump risk is **non-systematic** (diversifiable) $\Rightarrow$ model remains preference-free.

Strengths

- Analytical closed form; nests B&S ($\lambda = 0$)
- Explains volatility smile

Limitations

- Constant λ (no clustering)
- Symmetric log-normal jumps
- No stochastic volatility

Log-jump density

$$f_Y(y) = p \eta_1 e^{-\eta_1 y} 1_{y \geq 0} + q \eta_2 e^{\eta_2 y} 1_{y < 0}$$

- $p, q = 1 - p$: up/down probabilities; $1/\eta_i$: mean jump sizes
- $q > p, \eta_2 < \eta_1 \Rightarrow$ left skew (crashes sharper than rallies)

Why exponential?

Memoryless property $\Rightarrow$ overshoot tractable $\Rightarrow$ **closed forms for barrier & lookback options**

Improvements over Merton

- Asymmetric tails; matches leptokurtic returns
- Steeper downside implied-vol skew

Shared limitations

- Constant λ ; no stochastic volatility
- Incomplete market

Relevance to QQQ

Asymmetric structure fits the pronounced put skew of the NASDAQ-100.

Carr & Wu (2003): What Type of Process Underlies Options?

Core idea: read the process type from short-maturity option price decay.

Term Decay Rates (OTM options)

Process type	OTM decay
Purely continuous (PC)	$O(e^{-c/T})$
Pure jump (PJ)	$O(T)$
Combined (CJ)	$O(T)$

Method: plot $\ln(P/T)$ vs $\ln T$ (term decay plot). Flat OTM slope ≈ 0 signals jumps; ATM slope ≈ -0.5 signals a continuous component.

S&P 500 findings (1999–2000)

- OTM slopes $\approx 0 \Rightarrow$ jump component present
- ATM slope $\approx -0.41 \Rightarrow$ continuous component always present
- Jump component varies over time; correlation with VIX ≈ -0.58

Implication

Combined (CJ) models like Merton and Kou are empirically validated, but jump intensity must be **time-varying**.

Four nested models:

- **SV**: stochastic volatility only
- **SVJ**: SV + return jumps
- **SVCJ**: contemporaneous jumps in S and V
- **SVIJ**: independent jumps in S and V

SVCJ dynamics:

$$dV_t = \kappa(\theta - V_t) dt + \sigma_v \sqrt{V_{t-}} dW_t^V + \xi^v dN_t$$

$$dY_t = \mu dt + \sqrt{V_{t-}} dW_t^Y + \xi^y dN_t$$

Key empirical results

- Log-Bayes factor SV vs. SVIJ: **59.45** (S&P 500)
- 1987 crash: SV needs -8σ shock; SVCJ explains it with one jump
- Volatility jumps: persistent rise; return jumps: transient

MCMC estimation

Simultaneously recovers parameters *and* latent volatility, jump times, and jump sizes.

Nasdaq 100 finding

Jumps $\approx 3\times$ more frequent than in S&P — directly relevant to QQQ.

Hawkes intensity process:

$$d\lambda_t = \beta(\lambda_\infty - \lambda_t) dt + \alpha dN_t$$

- λ_∞ : background intensity
- α : self-excitation spike per jump
- β : exponential decay rate

Key feature: each jump raises λ_t by $\alpha \Rightarrow$ **jump clustering**

Log-price also includes stochastic variance V_t (CIR) and convenience yield δ_t (OU). Affine structure $\Rightarrow$ futures prices in closed form.

Why it matters

Standard Poisson: jump probability independent of past.
Hawkes: after a shock, further extreme moves are **more likely** for days/weeks.

Empirical results (WTI, 2008–2018)

- Log Bayes factor (Hawkes vs. constant): **9.13** $\Rightarrow$ decisive
- $\hat{\alpha} = 135.19$; confirms clustering
- Lower out-of-sample MAE for futures

For QQQ

Constant- λ models *underestimate* consecutive large moves after a first shock.

Problem: in incomplete jump markets, perfect replication is impossible.

Solution: minimise expected squared hedging error:

$$J(z, \pi) = \mathbb{E} \left[\frac{1}{2} (X^{z, \pi}(T) - F)^2 \right]$$

Stackelberg game:

- 1 Leader chooses initial capital z (option price)
- 2 Follower picks optimal portfolio $\hat{\pi}$
- 3 Result: $\hat{z} = \mathbb{E}^{Q^*}[F]$

Implementation: LSTM network approximates π_t at each time step.

Numerical results

Model	MV price	Merton
Black–Scholes	0.500	0.500
Merton	0.519	0.515
Kou	0.499	—

Key insight

Merton assumes jump risk is diversifiable. MV pricing **correctly prices the unhedgeable component**, yielding a symmetric error distribution vs. Merton's occasional large losses.

Advantage for Kou

No closed form exists $\Rightarrow$ LSTM is indispensable.

Physical measure $\mathbb{P}$:

$$\frac{dS_t}{S_{t-}} = (\mu - \lambda^{\mathbb{P}} \kappa^{\mathbb{P}}) dt + \sigma dW_t^{\mathbb{P}} + (e^J - 1) dN_t^{\mathbb{P}}$$

Risk-neutral measure $\mathbb{Q}$:

$$\frac{dS_t}{S_{t-}} = (r - \lambda \kappa) dt + \sigma dW_t^{\mathbb{Q}} + (e^J - 1) dN_t$$

Drift correction $\kappa = \mathbb{E}^{\mathbb{Q}}[e^J - 1]$ ensures $e^{-rt} S_t$ is a $\mathbb{Q}$ -martingale.

Why a jump risk premium exists

Girsanov fixes the Brownian drift uniquely, but the jump measure is **not uniquely determined** by no-arbitrage.

$\Rightarrow \lambda^{\mathbb{Q}} - \lambda^{\mathbb{P}}$ is a **priced jump risk premium**.

Characteristic function (Merton)

$$\phi_X(u) = \exp\left(iu\nu\tau - \frac{1}{2}\sigma^2 u^2 \tau + \lambda\tau(\Psi_J(u) - 1)\right)$$

$$\Psi_J(u) = e^{iu\mu_J - \frac{1}{2}\sigma_J^2 u^2} \quad (J \sim \mathcal{N})$$

Condition on n jumps:

$$C^{\text{Merton}} = \sum_{n=0}^{\infty} e^{-\lambda' \tau} \frac{(\lambda' \tau)^n}{n!} C^{\text{BS}}(S, K, \tau, r_n, \sigma_n)$$

where $\sigma_n^2 = \sigma^2 + n\sigma_J^2/\tau$, $\lambda' = \lambda(1 + \kappa)$

Incompleteness

At a jump of size J , the option changes by $C(Se^J) - C(S)$. A Δ -hedge changes by $\Delta \cdot S(e^J - 1)$. These can match for **all** J only if the payoff is linear.

$\Rightarrow$ Jump risk **cannot be perfectly hedged**.

Kou characteristic function

$$\psi_J^{\text{Kou}}(u) = \frac{p\eta_1}{\eta_1 - iu} + \frac{q\eta_2}{\eta_2 + iu}$$

Plugs directly into $\phi_X \Rightarrow$ COS engine unchanged.

Terminal log-price (exact):

$$X_T = X_0 + \nu T + \sigma W_T + \sum_{i=1}^{N_T} J_i$$

$$N_T \sim \text{Poisson}(\lambda T), \quad J_i \sim \mathcal{N}(\mu_J, \sigma_J^2)$$

Plain MC estimator:

$$\hat{C}_{\text{MC}} = \frac{1}{M} \sum_{m=1}^M e^{-rT} (S_T^{(m)} - K)^+$$

Three variance reduction methods

- 1 **Antithetic variates:** pair paths with Z and $-Z$
- 2 **Moment matching:** rescale $\bar{S}_T \rightarrow S_0 e^{rT}$
- 3 **B&S control variate:**

$$\hat{C}_{\text{CV}} = \bar{Y} - \hat{\beta}(\bar{X} - C^{\text{BS}})$$

Convergence

$\text{RMSE}(M) \propto M^{-1/2}$ validated against analytic Merton formula.

Fang & Oosterlee (2008) formula:

$$C_0 = e^{-rT} \sum_{k=0}^{N-1} \Re[\phi(u_k) e^{-iu_k a}] U_k$$

$$U_k = \frac{2}{b-a} [\chi_k(c, b) - K \psi_k(c, b)]$$

Helper integrals χ_k, ψ_k computed in closed form.

Truncation: $a = \kappa_1 - L\sqrt{\kappa_2 + \sqrt{|\kappa_4|}}$, $L = 10$.

Spectral convergence

N	$ C^{\text{COS}} - C^{\text{exact}} $
16	3.7×10^{-1}
32	2.0×10^{-3}
64	9.5×10^{-11}
128	2.8×10^{-13}

Key advantage

$N = 128$ reaches floating-point precision. Both Merton and Kou use the same COS engine — only ϕ changes.

Digital options

Cash-or-nothing and asset-or-nothing: swap U_k coefficients. Parity checks pass to machine precision.

Merton vs. Kou: Implied Volatility Smiles

Merton smile

- Approximately **symmetric** about ATM
- Log-normal jumps carry only mild negative skew
- Underfits the steep put wing of equity indices

Kou smile

- Pronounced **negative skew**
- Heavier mean down-jump ($1/\eta_2$) inflates OTM put IV
- Better match to QQQ option surface

Parameter effects on the surface

Parameter	Effect
σ	Level of IV
λ	Amount of tail risk priced
$\mu_J < 0$	Downside skew
σ_J	Tail thickness / kurtosis
p, η_1, η_2	Asymmetry (Kou only)

Calibration strategy (W6)

Fit Merton/Kou to QQQ option cross-section; compare $\lambda^{\mathbb{Q}}$ vs $\lambda^{\mathbb{P}}$ for the jump risk premium.

Sample: 2023-05-23 to 2026-05-20, $N = 732$ trading days.

Return distribution

Statistic	Value
Mean (daily)	0.000930
Std. dev	0.01242
Skewness	0.379
Excess kurtosis	10.59
JB p -value	$< 10^{-6}$
Min / Max return	-6.4% / +11.3%

Kurtosis = 10.59 $\gg 0$

Far heavier tails than the normal distribution. Directly motivates a jump component.

Option data (2026-05-20)

901 contracts across 6 expiries after cleaning.

Criteria: live bid/ask *or* stale last price with open interest.

Arbitrage flags: monotonicity + PCP checks.

Rolling 20-day quantities:

$$RV_t = \sum_{s \in \mathcal{W}_t} r_s^2, \quad BPV_t = \frac{1}{\mu_1^2} \sum |r_s| |r_{s-1}|$$

Log-ratio statistic:

$$\ell_t = \ln(RV_t / BPV_t), \quad Z_t = \frac{\ell_t - \bar{\ell}}{\hat{s}_\ell}$$

- Pure diffusion: $RV \approx BPV$, $Z_t \approx 0$
- Jump present: RV inflated, Z_t spikes
- Two-condition rule: $|Z_t| > 2$ **and** $|r_t| > 3\hat{\sigma}_t$

Detection results

Metric	Value
Single-condition days	99
Ann. rate (single)	34 / year
Two-condition days	4
Ann. rate (two-cond.)	1.38 / year
Literature benchmark	2–6 / year

Lognormal jump-size fit

$$\hat{\mu}_J = -4.89, \quad \hat{\sigma}_J = 1.16$$

KS test $p = 0.118$ — not rejected at 5%.

Broadly consistent with Merton assumption.

Major Jump Dates (Top 10 by $|Z_t|$)

Date	r_t	Z_t	2-cond.	Event
2026-01-20	−2.1%	4.09		Trump inauguration; tariff executive orders
2025-10-10	−3.5%	4.08	✓	US–China tariff escalation (second wave)
2024-08-30	+1.2%	3.72		Recovery from yen carry-trade unwind
2025-08-01	−2.0%	3.68	✓	Weak payrolls; Sahm Rule triggered
2023-07-20	−2.3%	3.64		Fed hawkishness; rate expectations reset
2024-12-18	−3.7%	3.40	✓	Fed hawkish pivot; fewer 2025 rate cuts
2025-04-03	−5.5%	3.33		“Liberation Day” tariff shock
2024-06-05	+2.0%	3.14		Strong ADP payrolls surprise
2023-07-25	+0.7%	3.00		Mixed tech earnings; elevated volatility
2026-04-28	−1.0%	2.87		Trade-policy uncertainty; tariff risk-off

Identified events align with macro shocks, tariff announcements, and Fed surprises — exactly the discontinuous news that Merton’s model captures.

Jump Risk Premium: The Central Economic Quantity

Connecting the pieces

Physical measure $\mathbb{P}$
 $\lambda^{\mathbb{P}}$ from time-series



Risk-neutral $\mathbb{Q}$
 $\lambda^{\mathbb{Q}}$ from options



Jump Risk Premium
 $\lambda^{\mathbb{Q}} - \lambda^{\mathbb{P}}$

Current estimates ($\lambda^{\mathbb{P}}$)

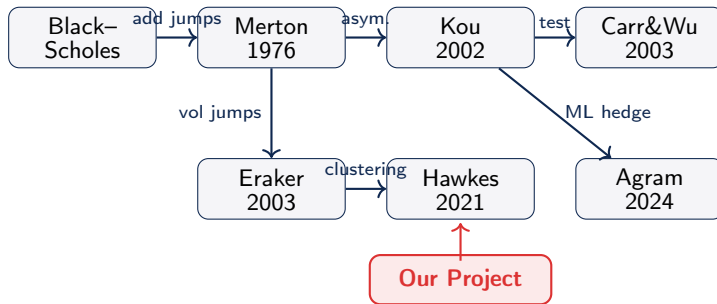
- Two-condition: 1.38 / year (lower bound)
- Single-condition: 34 / year (upper bound)
- Plausible structural range: 2–6 / year

W6 objective

Calibrate Merton and Kou to QQQ option surface to recover $\lambda^{\mathbb{Q}}$.

If $\lambda^{\mathbb{Q}} > \lambda^{\mathbb{P}}$: investors require compensation for bearing jump risk beyond pure expected value.

Literature Arc & Project Roadmap



W1: Lit. Review → **W2:** Math → **W3:** Monte Carlo → **W4:** COS → **W5:** Data & jump detection → **W6:** Calibration & risk premium

What we showed

- 1 B&S fails: fat tails + volatility smile
- 2 Merton (1976): Poisson jumps, closed-form pricing
- 3 Kou (2002): asymmetric jumps, steeper skew
- 4 Carr & Wu (2003): jumps confirmed in index options
- 5 Eraker et al. (2003): volatility jumps necessary
- 6 Hawkes: jump clustering modelled endogenously
- 7 Deep learning: LSTM hedges in incomplete markets

Technical achievements

- Monte Carlo with three variance-reduction methods
- COS engine: spectral convergence at $N = 128$
- 732 days QQQ data; excess kurtosis = 10.6
- Jump detection: 4 confirmed, 99 candidate days
- Lognormal fit consistent with Merton ($p = 0.12$)

Central finding

Empirical QQQ kurtosis and option term structure are consistent with a combined jump-diffusion. The gap $\lambda^{\mathbb{Q}} - \lambda^{\mathbb{P}}$ is the jump risk premium to be estimated in W6.

Thank You

Merton Jump-Diffusion: Pricing and Harvesting
the Equity Jump Risk Premium

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